Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.)

Program / Semester: **B. Tech (IV Sem)**	Branch: **Computer Science and Technology**
Subject: **Discrete Mathematical Structure**	Subject Code: **B141411(014)**
Total / Minimum-Pass Marks (End Semester Exam): **100/35**	L: 3 T: 1 P: 0 Credits: 4
Class Tests & Assignments to be conducted: 2 each	Duration (End Semester Exam): 03 Hours

Course Objectives

The objective of this course is to familiarize the prospective Computer science and IT engineers with techniques in discrete mathematics. It aims to equip the students with standard concepts and tools of logic that will serve them well towards tackling applications that they would find useful in their respective discipline.

UNIT-I MATHEMATICAL LOGIC & BOOLEAN ALGEBRA: Basic concept of mathematical logic, Statements, Connectives, Conditional and biconditional statements, Logical equivalence, Logical implication & quantifiers, Basic concept of Boolean Algebra, Properties of Boolean Algebra, Boolean functions, Disjunctive & conjunctive normal forms of Boolean functions, Applications of Boolean Algebra in switching circuits.

UNIT-II COMBINATORICS: Permutation and combination, Pigeon-hole principle, Mathematical induction, Principle of Inclusion and Exclusion, Generating function, Recurrence relation.

UNIT-III RELATIONS & FUNCTIONS: Relations, Properties of relations, Equivalence relation, Composition of relations, Introduction to function, Inverse, Identity, Injective, Surjective & Bijective functions, Composition of functions, Some special functions.

UNIT-IV GRAPH THEORY: Introduction to graphs, Types of graphs, Connectivity: Walks, Paths & Circuits, Eulerian and Hamiltonian graphs, Shortest path problems-Dijkstra Algorithm, Basic concept of tree: spanning tree, minimum spanning tree, prims Algorithm, Cut set.

UNIT-V ALGEBRAIC STRUCTURES: Binary operation on sets, Monoid, Semigroup and Groups: Finite and infinite groups, Subgroups, Cosets, theorem, Homomorphism, Isomorphism.

Text Books:

1. A Text Book of Discrete Mathematics, Swapan Kumar Sarkar S. Chand & Company
2. Elements of Discrete Mathematics C.L. Liu Tata McGraw-Hill
3. Discrete Mathematical Structures with Applications to Computer Science,J.P. Tremblay and R. Manohar Tata McGraw-Hill
4. Discrete Mathematical structures Bernard Kolman, Robert,C. Busby and Sharon,Cutler Ross,Pearson Education.

Reference Books:

1. Discrete Mathematics and Its Applications Kenneth H. Rosen McGraw Hill
2. Graph Theory with Applications to Engineering and Computer Science Narsingh Deo,Prentice Hall of India
3. Discrete Mathematics for Computer Scientists and Mathematicians, J.L. Mott, A. Kandel,and T.P. Baker Prentice Hall of India

Course Outcomes-On successful completion of the course, the student will be able to:

1. Apply mathematical logic and Boolean algebra in switching circuits & logic circuits.
2. Implement concepts of Combinatorics in practical problems.
3. Use the concept of relations and functions.
4. Use the graph theory in problems like finding shortest distance between two points.
5. Apply algebraic structures in various fields of Computer science and Information Technology.

Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.)

Program / Semester: **B. Tech (IV Sem)**	Branch: **Computer Science and Technology**
Subject: **Computer Organization & Architecture**	Subject Code: **B141412(022)**
Total / Minimum-Pass Marks (End Semester Exam): **100/35**	L: 2 T: 1 P: 0 Credits: 3
Class Tests & Assignments to be conducted: 2 each	Duration (End Semester Exam): 03 Hours

Course Objectives

Course objective is to make student learn the structure, function and characteristics of computer systems. To identify the elements of modern instructions sets and their impact on processor design.

UNIT I: Basic Structure of Computers: Functional units, Basic operational concepts, Bus structures Addressing modes, subroutines: parameter passing, Instruction formats, BASIC PROCESSING UNIT: bus architecture, Instruction Cycle, sequencing of control signals, Hardwired control, Micro programmed Control, microinstruction format.

UNIT II: Arithmetic: Number representations and their operations, Design of Fast Adders, Signed multiplication, Booth's Algorithm, bit-pair recoding, Integer Division, Floating point numbers and operations, guard bits and rounding.

UNIT III: The Memory System: Various technologies used in memory design, Memory Hierarchy: Main Memory, Auxiliary Memory, Associative Memory, Cache memory, Virtual Memory. Memory Management Hardware, Multi-module memories and Interleaving.

UNIT IV: Input/ Output Organization: Peripheral Devices, I/O interfaces I/O-mapped I/O and memory-mapped I/O, interrupts and interrupt handling mechanisms, vectored interrupts, synchronous vs. asynchronous data transfer, Direct Memory Access.

UNIT V: Pipeline and Vector Processing: Parallel Processing, Pipelining, Arithmetic pipeline, Instruction Pipeline, RISC Pipeline, Vector Processing, Array Processors.

Text Books:

1., Computer System Architecture, M. Marris Mano, PHI
2., **Computer Organization**, V.C.Hamacher,Z.G.Vranesic and S.G.Zaky, McGraw Hill
3., Computer Organization & Architecture, W. Stallings, Pearson Education India

Reference Books:

1., **Structured Computer Organization**, A.S.Tanenbaum, Pearson
2., **Computer Architecture & Organization**, J.P. Hayes, McGraw-Hill

Course Outcomes-On successful completion of the course, the student will be able to:

1. Identify the basic hardware components of a computer system and represent system design in appropriate formats; addressing modes, an instruction sets as per the system configuration requirements.
2. Familiarize themselves with binary and hexadecimal number systems including computer arithmetic.
3. Outline organization of various parts of a system memory hierarchy
4. Demonstrate computer architecture concepts related to design of modern processors, memories and I/O.
5. Basics functionality of systems: parallel, pipelined, superscalar and RISC/CISC architectures.

Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.)

Program / Semester: **B. Tech (IV Sem)**	Branch: **Computer Science and Technology**
Subject: **Java Programming**	Subject Code: **B141414(022)**
Total / Minimum-Pass Marks (End Semester Exam): **100/35**	L: 3 T: 1 P: 0 Credits: 4
Class Tests & Assignments to be conducted: 2 each	Duration (End Semester Exam): 03 Hours

Course Objectives

This course is designed to help students understand and apply the basic concepts of the Java programming language. It focuses on demonstrating the use of classes and exception handling for building robust applications, as well as implementing multithreading and file input/output operations for improved efficiency. Students will explore the use of AWT and Swing components for GUI development, along with database connectivity and networking features in Java. The course also highlights the effective use of the collections framework and generics to manage and process data in a flexible and type-safe manner.

UNIT- I Introduction: Introduction & Fundamentals of JAVA, Background of JAVA, About Java Technology, Java's architecture, Reading console inputs, Arrays, Constructors, Finalize method, final, this method and reference, static members. String class. Package concepts. Wrapper classes (Integer, Boolean, Character, etc.).

UNIT- II Class concepts & Exceptions: Concrete class, Abstract class, Interface, Inner classes. Aggregation, Composition and Inheritance, super method and reference. Method overloading and overriding. Singleton classes. Exception Handling: Built-in, User defined, Checked and Unchecked.

UNIT-III Multi-threading & IO: Multi-threading: Thread concept, Thread class, Runnable interface, Creating customized threads, Thread synchronization. Java I/O classes for disk files: File, FileInputStream, FileOutputStream, FileWriter, FileReader, ObjectInputStream, ObjectOutputStream.

UNIT-IV GUI, JDBC & Networking: AWT & SWING: Frame, Panel, Button, Dialog, CheckBox, Choice, List, JButton, JComboBox, JFrame, JPanel, JRadioButton, JScrollPane, JTabbedPane. Applets. JDBC: Type1 to Type4 drivers. Java Networking: ServerSocket, Socket.

UNIT-V Collections & Generics: Collections Frameworks: HashSet, TreeSet, ArrayList, LinkedList, Vector, HashMap, TreeMap, Hashtable classes. Generics in Java: Creating Generic methods and classes. Creating and running executable JAR (Java ARchives).

Text Books:

1. Java A Beginner's Guide, 7th edition, Herbert Schildt, Oracle Press
2. Java Generics and Collections, Maurice Naftalin, O'Reilly.

Reference Books:

1. Java in a Nutshell Benjamin J Evans O'Reilly
2. Head First Java Kathy Sierra O'Reilly.

Course Outcomes-On successful completion of the course, the student will be able to:

1. Apply the basics of Java programming language for developing applications.
2. Appreciate the components of Java programming for problem solving.
3. Apply the concept of classes, packages, exception handling, multi-threading, GUI programming and generics.

Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.)

Program / Semester: **B. Tech (IV Sem)**	Branch: **Computer Science and Technology**
Subject: **Database Management Systems**	Subject Code: **B141413(022)**
Total / Minimum-Pass Marks (End Semester Exam): **100/35**	L: 2 T: 1 P: 0 Credits: 3
Class Tests & Assignments to be conducted: 2 each	Duration (End Semester Exam): 03 Hours

Course Objectives

The objective of this course is to understand the role of a database management system, its users in an organization, basic database concepts, including the structure and operation of the relational data model and can successfully apply logical database design principles, including E-R diagrams and database normalization. This syllabus also aims to construct simple and moderately advanced database queries using Structured Query Language (SQL).

UNIT I: Database System Applications, database System VS file System, View of Data, Data Abstraction, Instances and Schemas, Data base Users and Administrator, Data base System Structure: Storage Manager, the Query Processor. Data base design and ER diagrams: Beyond ER Design Entities, Attributes and Entity sets, Relationships and Relationship sets, Additional features of ER Model, Concept Design with the ER Model, Conceptual Design for Large enterprises.

UNIT II: Introduction to the Relational Model, Integrity Constraint Over relations, Enforcing Integrity constraints, querying relational data, Logical data base Design, Introduction to Views, Destroying altering Tables and Views. Relational Algebra: Selection and projection set operations, renaming, Joins, Division, Examples of Algebra overviews, Relational calculus: Tuple relational Calculus, Domain relational calculus, Expressive Power of Algebra and Calculus**.**

UNIT III: Form of Basic SQL Query, Examples of Basic SQL Queries, Introduction to Nested Queries, Correlated Nested Queries, Set Comparison Operators, Aggregative Operators, NULL values, Comparison using Null values, Logical connectivity's: AND, OR and NOT, Impact on SQL Constructs, Outer Joins, Disallowing NULL values, Complex Integrity Constraints in SQL Triggers and Active Data bases.

UNIT IV: Schema refinement: Problems Caused by redundancy, Decompositions, Problem related to decomposition, reasoning about FDs, FIRST, SECOND, THIRD Normal forms, BCNF, Lossless join Decomposition, Dependency preserving Decomposition, Schema refinement in Data Base Design, Multi valued Dependencies, FORTH Normal Form.

UNIT V: Transaction Concept, Transaction State, Implementation of Atomicity and Durability, Concurrent , Executions, Serializability, Recoverability, Implementation of Isolation, Testing for serializability, Lock Based Protocols, Timestamp Based Protocols, Validation Based Protocols, Recovery and Atomicity, Log Based Recovery, Recovery with Concurrent Transactions, Indexing, Cluster Indexes, Primary and Secondary Indexes, Index data Structures, Tree base Indexing, B and B+ Trees: A Dynamic Index Structure.

Text Books:

1., Data base System Concepts, Silberschatz, Korth, Tata McGraw-Hill
2., Data base Management Systems**,** Raghurama Krishnan, Johannes Gehrke, TATA McGraw Hill

Reference Books:

1., Data base Systems design, Implementation, and Management, Peter Rob & Carlos Coronel,
2., Fundamentals of Database Systems, Elmasri Navathe, Pearson Education
3., Introduction to Database Systems, C.J. Date, Pearson Education

Course Outcomes-On successful completion of the course, the student will be able to:

1. Describe the basic concepts of RDBMS and relational data model.
2. Discuss the relational database theory & be able to write relational algebra expressions for queries.
3. Write queries using SQL by knowing the importance of data & its requirements in any applications.
4. Design Databases and improvement through normalization.
5. Identify basic issues of transaction, its processing and concurrency control and be familiar with basic database storage structures and access techniques, indexing methods.

Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.)

Program / Semester: **B. Tech (IV Sem)**	Branch: **Computer Science and Technology**
Subject: **Theory of Computation**	Subject Code: **B141415(022)**
Total / Minimum-Pass Marks (End Semester Exam): **100/35**	L: 2 T: 1 P: 0 Credits: 3
Class Tests & Assignments to be conducted: 2 each	Duration (End Semester Exam): 03 Hours

Course Objectives

This course focuses on the basic theory of computer science and formal methods of computation like automata theory, formal languages, grammars and Turing Machines. The objective of this course is to explore the theoretical foundations of computer science from the perspective of formal languages and classify machines by their power to recognize languages.

UNIT I: The Theory of Automata: Introduction to automata theory, Examples of automata machine, Finite automata as a language acceptor and translator, Deterministic finite automata. Non deterministic finite automata, finite automata with output (Mealy Machine. Moore machine), Finite automata with ε moves, Minimizing number of states of a DFA, My hill Nerode theorem, Properties and limitation of FSM, Application of finite automat.

UNIT II: Regular Expressions : Alphabet, String and Languages, Regular expression, Properties of Regular Expression, Finite automata and Regular expressions, Arden's Theorem, Regular Expression to DFA conversion & vice versa. Pumping lemma for regular sets, Application of pumping lemma, Regular sets and Regular grammar, Closure properties of regular sets. Decision algorithm for regular sets and regular grammar.

UNIT III: Grammars: Definition and types of grammar, Chomsky hierarchy of grammar, Relation between types of grammars, Context free grammar, Left most & right most derivation trees, Ambiguity in grammar, Simplification of context free grammar, Chomsky Normal From, Greibach Normal From, properties of context free language, Pumping lemma for context free language, Decision algorithm for context tree language

UNIT IV: Push Down Automata and Turing Machine: Basic definitions, Deterministic push down automata and non deterministic push down automata, Acceptance of push down automata, Push down automata and context free language, Turing machine model, Representation of Turing Machine, Construction of Turing Machine for simple problem's, Universal Turing machine and other modifications .Church's Hypothesis, , Halting problem of Turing Machine

UNIT V: Computability: Introduction and Basic concepts, Recursive function, Partial recursive function, Initial functions, Composition of functions, Ackerman's function, Recursively Enumerable and Recursive languages, Decidable and undecidable problem, Post correspondence problem, Space and time complexity.

Text Books:

1. Introduction to Automata Theory, Language and Computation, John E. Hopcroft and Jeffrey D. Ullman, Narosa Publishing House
2. Theory of Computer Science Automata Languages and Computation, K.L.P Mishra, N Chandrasekran, PHI
3. Introduction to Languages and the Theory of Computation, John Martin, Tata McGraw Hill.

Reference Books:

1. Introduction to Formal Languages Automata Theory and Computation, Kamala Krithivasan, Rama R., Pearson
2. Automata and Computability, Dexter C. Kozen, Springer
3. Elements of the Theory of Computation, Harry R. Lewis and Christos H. Papadimitriou, Pearson Education

Course Outcomes-On successful completion of the course, the student will be able to:

1. Design finite automata to accept a set of strings of a language.
2. Determine whether the given language is regular or not.
3. Design context free grammars to generate strings of context free language.
4. Design push down automata and the equivalent context free grammars and Design Turing machine.
5. Distinguish between computability and non computability, Decidability and undesirability.

Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.)

Program / Semester: **B. Tech (IV Sem)**	Branch: **Computer Science and Technology**
Subject: **Computer Organization & Architecture (Lab)**	Subject Code: **B141421(022)**
Total / Minimum-Pass Marks (End Semester Exam): **40/20**	L: 0 T: 0 P: 2 Credits: 1

Course objective is to understand the Architecture of computer system and its components. The students will learn maintenance, upgrading and repairing in various PC systems.

Experiment number 1 to 5 can be performed in C++ / Python / VHDL.

1.Write a program for IEEE-754 floating point representation and perform Addition/Subtraction.

2. Write a program for binary multiplication.

3. Write a program to perform signed bit multiplication using Booth's algorithm.

4. Write programs to simulate memory allocation policies

   a) First – fit algorithm

   b) Best- fit algorithm

5. Write a program to design basic ALU.

6. Assembling of a Personal Computer:

7. Learn Hard disk configuration and partitioning.

8. Study of Motherboard and its components.

9. Study of SMPS

10. Study of BIOS setup and Configuration

11. Study the functioning of peripheral devices outside the computer (keyboard, mouse, printer, scanner, monitor etc.)

Text Books:

1., Computer System Architecture, M. Marris Mano, PHI
2., Computer Organization, V.C.Hamacher,Z.G.Vranesic and S.G.Zaky, McGraw Hill
3., Computer Organization & Architecture, W. Stallings, Pearson Education India

Course Outcomes-On successful completion of the course, the student will be able to:

1. Familiarize themselves with binary and hexa decimal number systems including computer arithmetic.
2. Apply the organization of various parts of a system memory hierarchy.
3. Assemble the hardware part of the computer system and will be able to partition the memory and format the system.
4. Identify the various components of motherboard and SMPS.
5. Identify functioning of input /output peripheral device.

Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.)

Program / Semester: **B. Tech (IV Sem)**	Branch: **Computer Science and Technology**
Subject: **Database Management Systems (Lab)**	Subject Code: **B141422(022)**
Total / Minimum-Pass Marks (End Semester Exam): **40/20**	L: 0 T: 0 P: 2 Credits: 1

Course Objectives

The objective of this course is to understand the role of a database management system, its users in an organization, basic database concepts, including the structure and operation of the relational data model and can successfully apply logical database design principles, including E-R diagrams and database normalization. This syllabus also aims to construct simple and moderately advanced database queries using Structured Query Language (SQL).

List of Experiments: (At least Ten experiments are to be performed by each student)

Topics to be covered during LAB

1. How to install Oracle/ MySQL/ PostgreSQL in Windows/ Linux platforms. Creating users and database objects, inserting rows in the database tables.
2. Introduction to SQL, DDL, DML, DCL, TCL queries and constraints. Populating and manipulating database tables using DML statements, Indexes in SQL.
3. Selecting data from tables : SELECT statement, where clause, having clause, group by, order by, selecting NULL values, use of IN and DISTINCT keywords.
4. SQL functions: Study the use of SQL string, date, arithmetic and aggregate functions with examples.
5. JOINS : study the use of joining tables using natural, inner, outer joins.
6. Subqueries and set operations: Study the use of nested queries and how to apply them in database.
7. Views : Study the use of inline and external views.
8. Introduction to PL/ SQL : Pl/ SQL block, PL/ SQL statements, if else statements, looping statements.
9. Cursor : Study the use of cursor and exceptions.
10. Functions and procedures : Study the use of functions and procedures in PL/ SQL programs.
11. Trigger : Study the use of triggers to enforce constraints.
12. Forms and Report generation using PL/SQL.
13. Create a Database application using Oracle / Mysql/ PostgreSQL using any front end technologies.

Text Books:

1., Data base System Concepts, Silberschatz, Korth, Tata McGraw-Hill
2., Data base Management Systems**,** Raghurama Krishnan, Johannes Gehrke, TATA McGraw Hill

Reference Books:

1., Data base Systems Design, Implementation, and Management, Peter Rob & Carlos Coronel, Cengage Learning
2., Fundamentals of Database Systems, Elmasri Navathe, Pearson Education
3., Introduction to Database Systems, C.J. Date, Pearson Education

Course Outcomes-On successful completion of the course, the student will be able to:

1. Write efficient DB handling codes in DML, DDL and will be able to construct queries using SQL by knowing the importance of data & its requirements in any applications.
2. Write codes using efficient database storage structures and access techniques.
3. Write programs in PL/SQL using cursor, functions, triggers.
4. Write programs in PL/SQL, to generate the Report.
5. Design the database of any organisation with Front end and Back end with database connectivity.

Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.)

Program / Semester: **B. Tech (IV Sem)**	Branch: **Computer Science and Technology**
Subject: **Java Programming (Lab)**	Subject Code: **B141423(022)**
Total / Minimum-Pass Marks (End Semester Exam): **40/20**	L: 0 T: 0 P: 2 Credits: 1

Course Objectives

This course enables students to apply the basic concepts of the Java programming language in developing robust and efficient applications. It covers the use of classes, packages, and exception handling to build structured programs, as well as the implementation of multithreading and file input/output operations for enhanced performance. Students will also learn to work with AWT and Swing components for graphical user interface development, integrate database connectivity, and apply network programming concepts using ServerSocket and Socket. Additionally, the course emphasizes the use of the collections framework and generics to manage and manipulate data effectively.

1. Develop a Java application to demonstrate the use of reading inputs from console.
2. Develop a Java application to demonstrate the use of constructors, this reference and this method.
3. Develop a Java application to demonstrate the use of String class.
4. Develop a Java application to demonstrate the use of Multi-threading.
5. Develop a Java application to demonstrate the use of Java I/O for reading and writing text files.
6. Develop a Java application to demonstrate the use of AWT and SWING components.
7. Develop a Java application to demonstrate the use of database connectivity (JDBC).
8. Develop a Java application to demonstrate the use of Socket and ServerSocket for peer to peer communication.
9. Develop a Java application to demonstrate the use of Collections framework.
10. Develop a Java application to demonstrate the use of Generics.

Recommended Books:

1. Java A Beginner's Guide, 7th edition, Herbert Schildt, Oracle Press
2. Java Generics and Collections, Maurice Naftalin, O'Reilly.

Course Outcomes-On successful completion of the course, the student will be able to:

1. Apply the concepts related to the components of Java programming for developing real world applications.
2. Combine the concepts of classes, packages, exception handling, multi-threading, GUI programming, JDBC, RMI and generics to build real world applications.

Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.)

Program / Semester: **B. Tech (IV Sem)**	Branch: **Computer Science and Technology**
Subject: **Mini Project -I**	Subject Code: **B141424(022)**
Total / Minimum-Pass Marks (End Semester Exam): **40/20**	L: 0 T: 0 P: 2 Credits: 1

Course Objectives

The objective of this course is to improve student ability to analyze, design and solve complex engineering problems through pedagogies (Project Based Learning) that support them in developing these skills. The goal here is not to passively absorb and reiterate information; but rather to actively engage with the content, work through it with others, relate to it through an analysis, use modern tools and effectively solve problems with the corresponding knowledge gained.

The Process Followed to Maintain the Quality of Student Projects are:

a). Allotments of Projects
b). Project Identification
c). Continuous Monitoring
d). Evaluation

a). Allotment of Projects:

i. Students form their team (max four students) and submit their areas in which they would like to pursue their projects.
iii. Through meeting and deliberations students are allotted guide depending on their preference and maximum number of groups under a faculty is limited to three.

b). Identification of projects

Students are asked to formulate problem statement and state objectives of their project in consultation with the project guide.

c). Continuous Monitoring

i. Progress is continuously monitored by guide and instructions are given how to proceed further during their project periods as per time table.
ii. Students submit weekly progress report to the project in-charge after consultation with their project guide.

d). Evaluation

i. In order to evaluate projects two project seminars (assessment) are taken in which student's team present their project through presentations and demonstrate their work.
ii. Students are assessed on the basis of their technical skill implementation, use of modern tools, communication skill, team work, health, safety and ethical practices and relevance of the project.
iii. At the end of the semesters a report is submitted by the students and student's projects are finally evaluated by external examiner in end semester practical examination based on demonstration by students.

Course Outcomes-On successful completion of the course, the student will be able to:

1. Identify, discuss and justify the technical aspects of the chosen project with a comprehensive and systematic approach.
2. Reproduce, improve and refine technical aspects of engineering projects applying appropriate techniques, resources, and modern engineering and IT tools.
3. Work as an individual and as a member or leader in teams in development of technical projects.
4. Follow management principle and value health, safety and ethical practices during project.
5. Communicate and report effectively project related activities and findings.

Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.)

Program / Semester: **B. Tech (IV Sem)**	Branch: **Computer Science and Technology**
Subject: **Environmental Sciences (Constitution of India)**	Subject Code: **B000406(046)**
Total / Minimum-Pass Marks (End Semester Exam): **0/0**	L: 0 T: 0 P: 2 Credits: 0

Unit 1: The Constituent Assembly & The Constitution Of India: Historical Context of Constituent Assembly, Compositions & Functions, Critical Evaluation, Features of Indian Constitution, Preamble to the Constitution of India, Introduction to Fundamental Rights, Right to Equality, Right to Freedom, Constitutional Position of Some Democratic Rights, Right Against Exploitation, Right to Freedom of Religion, Right To Constitutional Remedies, Directive Principles

Unit 2: Organs Of The Government: The President of India, Powers and Functions of President, Emergency Powers and the Position of the President, Union Council of Ministers, Prime Minister, The Rajya Sabha, The Lok Sabha& Lok Sabha Speaker, Relation between Lok Sabha & Rajya Sabha

Unit 3: Indian Judiciary: The Structure and Organization of the Judiciary & the High Court, The Supreme Court, Role of The Supreme Court, Judicial Activism in India, Basic Structure Doctrine & PIL

Unit 4: Federalism & Decentralization: Legislative Procedures of the Parliament, Parliamentary Committees, Centre-State Legislative Relations, Centre-State Administrative Relations, Centre-State Financial Relations, The 5th & 6th Schedules

Unit 5: Indian Municipality and Gram Panchayats: Municipality-1 (History of Indian Municipality), Municipality-2 (Organization & Functions), Panchayat---1 (Idea of Panchayat), Organization and Powers of Panchayats in India

Text Books:

1. Durga Das Basu --- Introduction to the Constitution of India, 23rd Edition (Gurgaon; LexisNexis, 2018).
2. J.C.Johari -- The Constitution of India: A Politico-Legal Study (Greater Noida: Sterling Publishers Pvt. Ltd. 2013).
3. Himangshu Roy and M.P.Singh – Indian Political System, 4th Edition (Bengalaru; Pearson Education, 2018)
4. Vidya Bhushan &VishnooBhagwan--- Indian Administration (S. Chand, 2011)

Reference Books:

1. S.R.Maheswari --- Indian Administration (Orient Blackswan, 2001)
2. Dr. A.Avasthi& A.P. Avasthi --- Indian Administration (L.N. Agarwal Educational Publishing, 2017).
3. B. L. Fadia --- Indian Government and Politics (Sahitya a. Bhawan, 13th Revised Edition, 2017).
4. P.M.Bakshi – The Constitution of India (Prayagraj, UP; a. Universal Law Publishing, January, 2018)

Course Outcomes-On successful completion of the course, the student will be able to:

1. The citizens of India learn to abide by the laws of Indian Parliament and the judiciary.
2. Indians become aware of their fundamental rights and duties from the Constitution of India.